Linxuan Wang

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Research Summary

My research focuses on efficient LLMs, ranging from diffusion language models to speculative decoding. I develop pretraining methods that teach diffusion language models to self-correct, and tree-based speculative decoding methods that accelerate LLM inference.

Education

Purdue University — 2024 – 2029 (Expected)

Ph.D. in Statistics — *West Lafayette, IN, United States*

- Research fields: Diffusion Language Models, Speculative Decoding
- Advisors: Prof. Guang Lin & Prof. Qifan Song

Duke University — 2022 - 2024

M.S. in Statistics — *Durham, NC, United States*

- Research fields: Bayesian Nonparametrics, Markov Chain Monte Carlo, Quantile Regression
- Advisor: Prof. Surya Tokdar

Wuhan University — 2018 - 2022

B.S. in Mathematics and B.A. in Economics — *Wuhan, Hubei, China*

- GPA: 3.8/4.0, Awards: Honor Graduate of Hongyi Honor College

Publications & Preprints

*(★ denotes selected work. * denotes equal contribution.)*

★ Liang, C.*, **Wang, L.***, Peng, X.*, Liu, H., Chen, Y., Liu, G., Lin, G., Song, Q., Li, J. (2026) UBTree: Parallel Tree Drafting via Unigram and Bigram Models for Speculative Decoding, [Paper][HuggingFace].

Bai, Y., Lu, B., Liu, Y., Akbari, E., Kolouri, S., **Wang, L.**, He, P., Wang, S., Zhang, R., Lin, G. (2026) SE(3)-MeanFlow: Few-Step Protein Backbone Generation on Lie Groups, *arXiv preprint*, [Paper].

Liang, J., **Wang, L.**, Lin, G., Song, Q. (2026) Breaking the $\sqrt{d}$ Communication Barrier in Federated Sampling with Adaptive Hamiltonian Monte Carlo, *NeurIPS 2026*.

★ **Wang, L.***, Wang, Z.*, Bai, Y.*, Deng, W., Lin, G., Song, Q. (2026) Generalized Discrete Diffusion with Self-Correction, *ICML 2026*, [Paper][Code][Project Page][NVIDIA NeMo AutoModel].

Wang, L. (2024) Bayesian Sparse Learning of Joint Quantile Planes Estimation. Master's thesis, Duke University, [Paper].

Research Experiences

Fast Generation in Diffusion Models — July 2025 - present

- Developed Self-Correcting Discrete Diffusion (SCDD), the **first ELBO-based pretraining framework for diffusion LMs to learn to self-correct without remasking**.
- Improved parallel decoding efficiency at GPT-2 scale. Integrated into NVIDIA NeMo AutoModel.

Bayesian Sparse Learning of Joint Quantile Regression — Jan 2023 - Mar 2024

- Developed Bayesian sparse learning methods for joint quantile regression, capturing local and global sparsity.
- Implemented an MCMC sampler in C and evaluated the method with simulations in R and a biomedical case study.

Work Experiences

Research Intern, Ant Group — May 2026 - Sep 2026

- Inclusion AI LLaDA Team, mentored by Dr. Jianguo Li; researched diffusion language models and speculative decoding.
- Co-led the **UBTree** project: parallelized candidate-transition scoring with a bigram selector and developed **tree-native** training using high-temperature rollouts and renormalized KL supervision to improve branch diversity.
- Achieved **5.84–6.94× average speedup** on Qwen3-4B/8B; outperformed DARTree in all 28 benchmark comparisons.
- Outperformed **DSpark** in SGLang: 5.36×/4.78× average speedups over autoregressive decoding on Ling3-Flash-124B/Qwen3.8-27B, respectively, representing **19.1%/10.6% higher speedups than DSpark**.

Academic Service

Conference Reviewer: ICLR 2027.